Exercises Session 1

Universidad Icesi

2 de febrero de 2026

1. Regular languages and automata theory

Exercise 1 *Given* $\Sigma = \{a, b, c, d\}$ *define* Σ^*

Exercise 2 *Given* $u \in \Sigma^*$ *and* $n \in \mathbb{N}$, *give a recursive definition for* u^n.

Exercise 3 *Given* $u \in \Sigma^*$ *and* $a \in \Sigma$ *give a recursive definition for* $|u|$.

Exercise 4 *Given* $u \in \Sigma^*$ *and* $a \in \Sigma$ *give a recursive definiton for* u^R.

Exercise 5 *If* $u, v \in \Sigma^*$, *show that* $(uv)^R = v^R u^R$.

Exercise 6 *Given an example of an alphabet* Σ *and two different languages* A, B *over* Σ *such that* $AB = BA$.

Exercise 7 *Show that* $A \cdot (B \cdot C) = (A \cdot B) \cdot C$.

Exercise 8 *Show that* $A \cdot \{\lambda\} = \{\lambda\} \cdot A = A$.

Exercise 9 *Show that* $A \cdot \emptyset = \emptyset \cdot A = \emptyset$.